Use the NIST cybersecurity framework to respond to a security incident

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an **incoming flood of ICMP packets**. **Normal internal network traffic could not access any network resources**. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that **a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall**. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets
- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

As a cybersecurity analyst, you are tasked with using this security event to create a plan to improve your company's network security, following the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF).

Incident report analysis

Summary	A Denial-of-Service (DoS) attack occurred when a malicious actor flooded the company's network with incoming ICMP packets through an unconfigured firewall. The attack overwhelmed network resources, preventing normal internal network traffic from accessing network resources. The incident disrupted the company's internal network for approximately two hours. The investigation identified an unconfigured firewall as the security weakness that allowed the attacker to send excessive ICMP traffic into the network. The incident management team responded by blocking incoming ICMP packets, taking non-critical network services offline and restoring critical network services. Following the incident, the network security team introduced additional security controls, including ICMP rate limiting, source IP address verification, network monitoring software and an Intrusion Detection and Prevention System (IDS/IPS).
Identify	The organisation should conduct regular security audits of its internal networks, systems, devices and access controls to identify vulnerabilities that could expose it to future attacks. The investigation identified an unconfigured firewall as the primary security weakness that allowed excessive incoming ICMP traffic to reach the company's network. This vulnerability enabled a malicious actor to overwhelm network resources, disrupting normal operations for approximately two hours. The organisation should review firewall configurations, identify other misconfigured network devices and assess existing security policies. Regular vulnerability assessments and configuration reviews should be

	introduced to identify and address security gaps before they can be exploited.
Protect	**To reduce the risk of similar attacks occurring in the future, the network security team implemented several protective measures:** ● ICMP rate limiting: Introduced firewall rules to restrict the number of incoming ICMP packets within a given period, reducing the risk of network resources being overwhelmed. ● Source IP address verification: Implemented firewall checks to identify potentially spoofed source IP addresses and prevent malicious traffic from entering the network. ● Network monitoring: Introduced monitoring software to identify unusual traffic patterns and potential security threats. ● IDS/IPS implementation: Deployed Intrusion Detection and Prevention System controls to identify and filter suspicious ICMP traffic based on predefined security rules. The organisation should also establish regular firewall configuration reviews and ensure network security policies are maintained and followed.
Detect	A Security Information and Event Management (SIEM) system should be used to collect and analyse firewall logs, network traffic data and security alerts. Network analysis tools such as Wireshark can help analysts inspect packet captures and identify unusual ICMP traffic patterns. IDS/IPS systems should be configured to detect suspicious traffic volumes, repeated ICMP requests and potential IP spoofing. Network traffic baselines should also be established so that unusual activity can be identified quickly. Regularly reviewing firewall logs, network traffic captures and security alerts will help the cybersecurity team identify potential DoS attacks early

	and reduce the time taken to detect future incidents.
Respond	The organisation should develop and regularly test an incident response plan specifically addressing DoS attacks. When an ICMP flood is detected, the security team should investigate the source, volume and characteristics of the incoming traffic. Appropriate firewall rules should be applied to rate-limit or block malicious packets, while IDS/IPS controls can help filter suspicious traffic. If necessary, non-critical network services should be temporarily disabled to preserve resources for essential business operations. The incident management team and relevant network administrators should be notified, and all response actions should be documented. Firewall logs and packet captures should be analysed to establish how the attack occurred, identify the security weaknesses involved and assess the effectiveness of the response. Following containment, the organisation should review the incident and introduce improvements to its security controls and response procedures.
Recover	Once the attack has been contained, critical network services should be restored in a controlled manner. The cybersecurity team should verify that internal users can access network resources and that essential business services are operating normally. Network performance should be tested, and traffic should be monitored closely to confirm that the ICMP flood has stopped and that no further suspicious activity is occurring. Firewall rules and IDS/IPS configurations should also be reviewed to ensure that the implemented security controls are functioning correctly. The organisation should document all recovery actions, communicate the

	restoration of services to employees and relevant stakeholders, and conduct a post-incident review. Lessons learned should be used to improve the incident response plan and reduce the likelihood and impact of similar attacks in the future.

Reflections/Notes: This incident demonstrates how a misconfigured firewall can expose an organisation to network-based attacks and disrupt business operations.

Applying the NIST Cybersecurity Framework helps an organisation approach cybersecurity incidents systematically, from identifying vulnerabilities and implementing protective measures to detecting threats, responding effectively and restoring normal operations.